Saurabh Dhamne · Prognostics and health management case study · Wind energy

Wind turbine normal-behaviour model, underperformance detection and short-term forecasting

1. Problem & domain

A turbine loses revenue silently. Stops are easy to see, but a turbine that keeps running a few percent below its potential (soiling, pitch offset, icing, sensor drift) never trips an alarm. On this 3.6 MW turbine a 5% loss costs ~$2,000/month at the 7.3 USc/kWh Turkish wind tariff. Operators compare against the manufacturer curve, but it **over-predicts this turbine by ~10%**: a 90%-of-expected rule flags 34% of healthy records, so alarms get ignored. **The opportunity:** learn what a healthy turbine should produce (a normal-behaviour model, NBM), then use it to (a) catch subtle losses early and (b) forecast output. **Why classical ML:** the data is structured 10-min sensor data governed by a physical law ($P \propto \tfrac{1}{2}\rho A C_p v^3$). The questions are numeric, and LLMs add nothing to that. A GenAI layer (alarm → work order) belongs on top only once the alarm is reliable. **Data:** Kaggle Wind Turbine SCADA, one turbine in Turkey, 2018, 50,530 rows. Real messiness: 3.9% missing, 4.2% stops, a derating plateau, a ~15% seasonal air-density effect. Limits: one turbine-year, and no temperature, status codes or fault labels.

2. Approach & algorithm decisions (each with the alternative rejected)

- **Chronological split** (train Jan–Aug, validation Sep–Oct, test Nov–Dec). **Every choice is made on validation.** Random split rejected: lag-1 autocorrelation is 0.98, so it would test on near-copies of training rows.
- **Normal-operation filter** before training the NBM: physical state rules, then per-0.5 m/s-bin median ± 3·MAD (IEC bins). Mean/std rejected, because the derates being removed would inflate the std and hide themselves.
- **NBM = monotone LightGBM on wind speed, turbulence and direction.** Compared against the manufacturer curve, the IEC binned curve and a 4-parameter logistic. Rejected: NN/LSTM (25k rows, 4 tabular features; a sequence input would let past power leak in), Gaussian process ($O(n^3)$); kept as the upgrade if intervals are needed), random forest (no monotone constraints). **No lagged power**, or the model copies the fault and the residual vanishes. The monotone constraint is accuracy-neutral (1.795 vs 1.807% val) but guarantees a physical curve where data is thin, where a dip would become a false alarm.
- **Feature ablation (validation):** direction and turbulence help, hour does not. **Day-of-year hurts on validation but would help on test.** Switching would be test leakage, so I did not. The real fix is the missing variable (temperature → air-density correction).
- **Forecast:** direct per-horizon models (recursive rejected: errors compound). A LightGBM predicting the power *level* lost to persistence, so it predicts the **change** P(t+h)–P(t), making persistence its starting point. Ridge AR tested and rejected (cannot saturate at rated power).
- **Detection:** residual → z per 1 m/s bin (noise is heteroscedastic) → one-sided **CUSUM**, the classical test for a small persistent shift. Threshold h set for ≤**1 false alarm/turbine/month** on healthy validation data, calibrated empirically because residuals are autocorrelated. k = 0.25 chosen by injecting faults into validation. Rejected: point thresholds (noise ≈ fault size), Isolation Forest (tested, below), and a rolling bias correction (tested: absorbs the fault, 5% detection 68% → 40%).

3. Results & error analysis (held-out test, Nov–Dec)

NBM (test, normal)	nMAE	Forecast, test nMAE	Persist.	LightGBM
Manufacturer curve	3.27%	10 min	3.49%	3.48%
IEC binned curve	2.46%	1 h	7.94%	7.86%
4-param. logistic	3.10%	3 h	13.01%	12.79%
LightGBM, monotone	**2.20%**	6 h	18.69%	18.04%

7-day injected fault, 40 trials	CUSUM	Median delay	Iso. Forest
Efficiency loss 3% / 5%	42% / 78%	52 h / 79 h	0% / 0%
Efficiency loss 10%	90%	52 h	0%
Anemometer drift +5%	98%	36 h	55%

- Faults are injected only below rated wind (above it, the pitch system recovers the loss). A detection is credited only if it comes **before** any alarm on the clean data. Isolation Forest fails because a 5% loss keeps every point inside the normal cloud.

- On clean test data, CUSUM flagged **three winter episodes 7–16 h before the turbine stopped** (consistent with icing; unconfirmed without temperature). One alarm (19 Nov) was a ~3% shortfall at a derating setpoint: real, but likely intended.
- **Where it fails:** (1) seasonal density: mean residual (actual – expected) swings –32 to +88 kW, masking faults in winter; (2) anemometer drift looks identical to real loss; (3) large sudden losses take ~8 h, because CUSUM is tuned for small shifts; (4) forecast gains are small and the 1-h gain wins only 5/9 backtest months. Without NWP weather input, persistence is near the ceiling.

4. Production & limitations

Production consideration: score each turbine every 10 min as a stateless service (model + per-bin scales + CUSUM state per turbine), with alarms into the maintenance queue. Monitor two health signals: monthly NBM bias on filter-normal data (retrain if |bias| > 1.5% of rated), and rolling 7-day forecast skill (fall back to persistence automatically if < 0). Retrain quarterly, only on periods signed off as healthy, so slow degradation is never learned as normal. **Limitation before real deployment:** no air density. Add temperature and pressure (IEC $v \cdot (\rho/\rho_0)^{1/3}$ correction), and add a neighbour-turbine wind cross-check to separate sensor drift from real loss. **Value:** 74 MWh (~$5.4k) lost in stops and 93 MWh (~$6.8k) in abnormal running in two months on one turbine. Catching a 5% loss in ~3 days instead of at a monthly review saves ~90% of it, about $90k/year for a 50-turbine farm with one such fault per turbine per year.

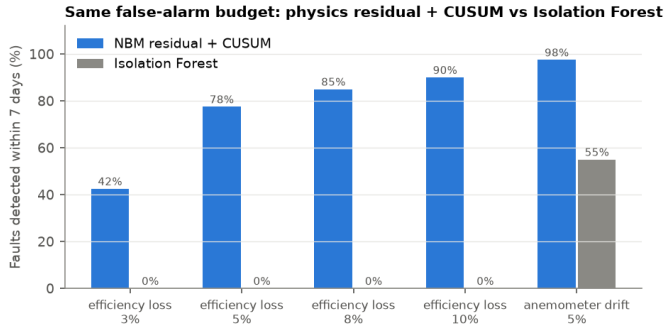


Fig. 1. Same false-alarm budget: physics residual + CUSUM vs Isolation Forest on injected faults.

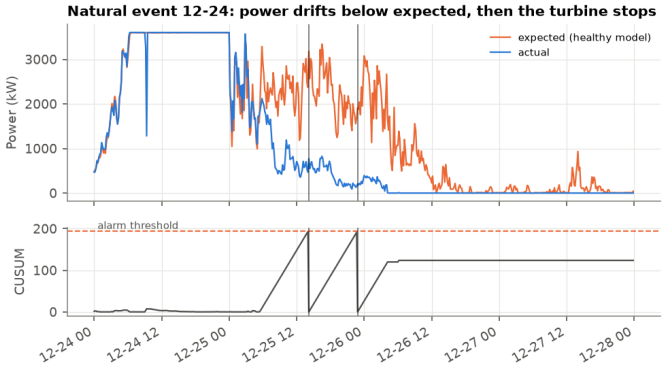


Fig. 2. Natural event, 25 Dec: power slides below expected; CUSUM alarms ~16 h before the stop.